



Effect of pyrolysis temperature and molecular weight on characterization of biochar derived dissolved organic matter from invasive plant and binding behavior with the selected pharmaceuticals[☆]

Wangyu Wang¹, Minghua Nie¹, Caixia Yan^{*}, Yulong Yuan, Aoxue Xu, Mingjun Ding, Peng Wang, Min Ju

School of Geography and Environment, Key Laboratory of Poyang Lake Wetland and Watershed Research, Ministry of Education, Jiangxi Normal University, 99 Ziyang Road, Nanchang, 330022, China

ARTICLE INFO

Keywords:

BDOM
Pharmaceutical
Binding
Molecular weight
Fluorescence

ABSTRACT

A comprehensive understanding of the characteristics of biochar released-dissolved organic matter (BDOM) derived from an invasive plant and its impact on the binding behavior of pharmaceuticals is essential for the application of biochar, yet has received less attention. In this study, the binding behavior of BDOM pyrolyzed at 300–700 °C with sulfathiazole, acetaminophen, chloramphenicol (CAP), and carbamazepine (CMZ) was investigated based on a multi-analytical approach. Generally, the pyrolysis temperature exhibited a more significant impact on the spectral properties of BDOM and pharmaceutical binding behavior than those of the molecular weight. With increased pyrolysis temperature, the dissolved organic carbon decreased while the proportion of the protein-like substance increased. The highest binding capacity towards the drugs was observed for the BDOM pyrolyzed at 500 °C with the molecular weight larger than 0.3 kDa. Moreover, the protein-like substance exhibited higher susceptible and released preferentially during the dialysis process and also showed more sensitivity and bound precedingly with the pharmaceuticals. The active binding points were the aliphatic C–OH, amide II N–H, carboxyl C=O, and phenolic-OH on the tryptophan-like substance. Furthermore, the binding affinity of the BDOM pyrolyzed at 500 °C was relatively high with the stability constant ($\log K_M$) of 4.51 ± 0.52 .

1. Introduction

Biochar, a carbon-rich porous material produced by the thermochemical conversion of biomass, is widely applied in soil amendment, carbon sequestration, and reduction of greenhouse gas emissions (He et al., 2021b; Wang et al., 2020b). It could be attributed to its low cost and unique physicochemical properties, such as a large specific surface area, ample functional surface groups, high cation/anion exchange capacity, long-term stability, etc. (Shaheen et al., 2019; Zhang et al., 2021b). Once biochar enters the environment, the biochar-derived dissolved organic matter (BDOM) is inevitably released, which is the most active and mobile component (Joseph et al., 2010).

BDOM is a readily soluble and heterogeneous mixture of organic compounds containing highly redox activity (Yang et al., 2020b). As accounting for 10% of the natural dissolved organic carbon in global

river fluxes, a considerable portion of the black carbon is BDOM (Jaffé et al., 2013). The abundance of nutrients (C, N, and P) provided by BDOM could help plant growth (Zhang et al., 2020a). Besides, BDOM also showed toxicity to some alga (e.g., *Chlorella* sp.) and suppressed seed germination (e.g., rice, corn, and wheat) due to its phenols, polycyclic aromatic hydrocarbons, and organic acids (Hao et al., 2018; Zhou et al., 2023). Moreover, the fate of BDOM in the environment can influence the transport and fate of pollutants in the environment, thus affecting the remediation effect of biochar (Yang et al., 2023). Therefore, a comprehensive understanding of the characteristics of BDOM and its impact on the environmental behavior of contaminants is essential for the application of biochar and ecosystem safety.

Conventionally, BDOM will be released from biochar under various aging processes (physical weathering, microbial decomposition, chemical oxidation, etc.) (Sun et al., 2021). However, most of the previous

[☆] This paper has been recommended for acceptance by Dr Amit Bhatnagar.

^{*} Corresponding author.

E-mail address: yancaixia@jxnu.edu.cn (C. Yan).

¹ These authors equally contributed to this work.

studies overall had an emphasis on the characteristics of the bulk BDOM. It has been shown that the composition of BDOM was size-dependent and the fate of many pollutants highly relied on the molecular weight of BDOM (Yu et al., 2023; Zhang et al., 2021a). Besides, the fluorescent characteristics of BDOM could be used as a fingerprint to trace the biochar migration in the field owing to the specific fluorescent components of BDOM (Jiang et al., 2022). Additionally, compared to the natural dissolved organic matter, BDOM contains rich condensed carbon components and oxygen-containing functional groups (e.g., carboxyl, hydroxyl, quinone, and phenolic groups), whose chemical composition and structure are important factors in determining the environmental transformation of pollutants (Spokas et al., 2014; Zhang et al., 2023). For instance, BDOM can directly affect the migration and transformation of heavy metals through hydrogen bonding, ligand, cation bond bridges, and complexation, which would change their sorption behavior to the other environmental sorbents (Huang et al., 2020; Huang et al., 2019). And although few, some of them had paid attention to the impact of the fluorescence substance (e.g., the protein-like substance), organic components (e.g., lipid/aliphatic), and functional groups (e.g., carboxylic group) on the BDOM-heavy metals binding behavior and simultaneously considered the influence of the pyrolysis temperature (Guo et al., 2022; Wu et al., 2022; Yan et al., 2023). Nonetheless, the study comprehensively with respect to the effect of the pyrolysis temperature on characterizing both different molecular weights of BDOM and these BDOMs binding on the organic pollutants (e.g., pharmaceuticals) has never been investigated. As an important emerging pollutant, the study on pharmaceuticals has received more and more attention on their treatment and environmental behavior, such as the removal of pharmaceuticals by the sorption of biochar (Ihsanullah et al., 2022). Thus, the impact of the released BDOM on the removal efficiency of pharmaceuticals by biochar would help to understand and improve this process.

Furthermore, a series of studies have demonstrated that the components and structures of BDOM were mainly influenced by the feedstock, pyrolysis temperature, and extraction protocol (Liu et al., 2023; Wei et al., 2019). With regards to the feedstock, the BDOM released from manure and herbaceous biochar was significantly more than that from woody biochar. A higher content of the multivalent metals could increase the thermal stability of biomass which leads to generating a large amount of dissolved organic carbon (DOC) in the manure-derived BDOM. Besides, the woody feedstocks usually contained more stable lignin than manure and herbaceous feedstocks (Liu et al., 2019; Wei et al., 2019). Compared to the feedstocks, a more significant impact on the spectral properties of BDOM and its binding to heavy metal was related to the pyrolysis temperature (Yan et al., 2023). Yet a study also showed that the feedstocks and pyrolysis temperature impacted different components of BDOM (i.e., the humic-like materials → temperature; the protein-like materials → feedstocks) (Rajapaksha et al., 2019). In general, the higher pyrolysis temperature resulted in a smaller molecular weight, less oxygen-containing functional group, higher condensed polycyclic aromatics, and lower DOC content of BDOM (Li et al., 2022). Furthermore, the amount of extracted BDOM increased with the increasing pH of the solution, whereas the effect of solution salinity on the quality and quantity of BDOM differed from the type of biochar (Liu et al., 2023). However, these studies mainly concerned the BDOM obtained from animal manure, wood chips, agricultural waste crops, and other materials, and less focus was placed on invasive plants (Dong et al., 2014; Poerschmann et al., 2015; Wei et al., 2019). Because of their strong adaptability, fast reproduction, and spreading, invasive plants have caused serious damage to the ecosystem and even affected the human living environment (Paini et al., 2016). As a high diversity and wide distribution of raw material, invasive plants pyrolyzed into biochar used in environmental remediation and soil amendment have been consequently and gradually considered an efficient resource utilization way to eradicate the plants (Feng et al., 2021). It has proved that biochar feedstocks from invasive plants could be an excellent absorbent for both inorganic (e.g., heavy metal) and organic pollutants (e.g.,

antibiotics) (Lian et al., 2020; Wang et al., 2023). Whereas, the relevant study on the characteristics of BDOM derived from the invasive plants and its binding behavior of pollutants (e.g., pharmaceuticals) is much scarcer so far.

Understanding the structure and material composition of BDOM and its binding mechanism to pollutants requires analysis using a hyphenated multi-method approach. For example, UV-visible absorption (UV-vis), synchronous fluorescence spectra (SFS) combined two-dimensional correlation spectroscopy (2D-COS), and three-dimensional excitation-emission matrix (3D-EEM) fluorescence spectroscopy combined parallel factor analysis (PARAFAC) is often applied to elucidate changes in the structure and composition of BDOM, as well as to characterize the mechanism of contaminant-fluorophore interactions due to their high efficiency, rapidity, and sensitivity (Wu et al., 2022; Yan et al., 2023).

Hence, four drugs with crescent hydrophobicity (i.e., ST, ACE, CAP, and CMZ) were selected in this study. BDOM was prepared using an invasive plant (*Alternanthera philoxeroides*) which was pyrolyzed at temperatures ranging from 300 °C to 700 °C. The binding capacity of BDOM with different molecular weights towards the four drugs was evaluated by dialysis equilibrium experiment. The binding mechanism between the fluorescent components in BDOM and the drugs was investigated by a fluorescence quenching method. The objectives of this study are to (1) investigate the effects of pyrolysis temperature and BDOM molecular weight on the spectral properties of BDOM; (2) explore the binding capacity of BDOM to the four drugs; (3) elucidate the binding mechanism of BDOM vs drugs, including binding site, binding substance, binding order, and binding affinity.

2. Materials and methods

2.1. Chemicals & materials

Chloramphenicol (CAP) was procured from J&K (purity ≥99%). Sulfathiazole (ST), acetaminophen (ACE), and **carbamazepine (CMZ)** were obtained from Sinopharm Chemical Reagents Ltd (purity ≥99%). The physicochemical properties of the four drugs are shown in Table S1. These four drugs (i.e., ST, ACE, CAP, and CMZ) have different hydrophobicity with an increasing value of $\log K_{ow}$ and belong to different types of drugs. The detailed description including their pollution condition and toxic effect is shown in Text S1. Methanol (MeOH, HPLC grade) was supplied by CNW (Germany). The chemical reagents used were of analytical grade and no further purification was required for their use. In addition, the stock solutions of chemical reagents in the experiments were prepared using Milli-Q water ($18.25 \text{ M } \Omega \text{ cm}^{-1}$) or methanol.

2.2. Sample collection and treatment

Alternanthera philoxeroides is one of the world's worst topical aquatic weeds, which has invaded most continents (except Africa and Europe) (Ye et al., 2003). It has been found in more than 20 provinces and regions of China, and become the most of weed nationwide (Wang et al., 2005). The *Alternanthera philoxeroides* sample was collected in December 2021 from the lakeside of a city lake (Yao Lake) belonging to Poyang Lake (the biggest fresh lake in China). Once transported to the laboratory, all collected biomass was washed with tap water to remove the debris and soil. Then, it was cut into a small piece of 2 cm sections, and dried in an oven at 60 °C. Before pyrolysis, the dried samples were crushed using a micro plant grinder. After that, the biomass powder was loaded on a ceramic ark gently compacted, and then placed in a tube furnace (BTF-1200 C) for pyrolysis carbonization. The biochar was prepared according to the previous research (Wang et al., 2020a). Firstly, an oxygen-limited state was created inside the device by feeding N_2 . Then, the temperature was increased at a rate of $10 \text{ }^\circ\text{C min}^{-1}$. After reaching the preset temperatures (300 °C, 400 °C, 500 °C, 600 °C and

700 °C), the pyrolysis temperature was maintained for 3 h. Finally, when cooling to room temperature, the samples were ground, sieved (100 mesh), and stored in self-sealing bags.

2.3. BDOM extraction

BDOM extraction was proceeded by the mixing of biochar and Milli-Q water at the ratio of 1:100 excluding the BDOM-500 °C used in the fluorescence quenching titration experiment was extracted at the ratio of 1:25. The mixture was shaken at 25 °C and 180 r min⁻¹ for 48 h in the dark. At the end of shaking, the sample was filtrated using a 0.45 µm pore size filter membrane and the filtrate obtained was BDOM. At the same time, the prepared BDOM solution was frozen and dried to characterize the apparent morphology and the functional groups. The BDOM prepared at the pyrolysis temperatures from 300 °C to 700 °C were recorded as BD3, BD4, BD5, BD6, and BD7, respectively. The extracted BDOM pyrolyzed at different temperatures was used to detect the spectrum and investigate the binding capacity of the pharmaceuticals.

2.4. Dialysis equilibrium experiment

The dialysis equilibrium experimental design is shown in Fig. 1. Aliquots of BDOM solution prepared at different pyrolysis temperatures were isolated into different molecular weights using the dialysis bags which were sealed at both ends. Four specifications of the dialysis bags were selected with the cut-off molecular weights of 0.3 kDa, 1 kDa, 3 kDa, and 10 kDa. Before loading the BDOM samples, the dialysis bags were pretreated (Text S2) and thoroughly washed with Milli-Q water to remove the excess storage solution. Besides, BDOM samples were diluted with different multiple according to the DOC concentration of BDOM pyrolyzed at different temperatures to avoid flocculation. Simultaneously, the mixture of the four drugs was spiked into 100 mL diluted BDOM samples with each final concentration of the drug at 1 mg L⁻¹. Then, these dialysis bags containing BDOM samples were placed in a beaker with 1 L Milli-Q water to isolate the BDOM samples smaller and larger than the corresponding molecular weights of the dialysis bag. This procedure was conducted on a magnetic stirrer for 48 h. At the interval times (0, 2, 4, 8, 12, 24, and 48 h), 2 mL samples were taken from the beakers (outside the dialysis bags) to analyze the concentration of the drugs for the smaller molecular weight of BDOM, as well as the

corresponding spectroscopy. Besides, these items in the dialysis bags were also detected at the end of the equilibrium time for the larger molecular weight of BDOM (Rizzuto et al., 2021). Pharmaceutical concentrations were analyzed by an Agilent 1260 prime infinity II ultra-performance liquid chromatography coupled to an Agilent Ultivo triple quadrupole mass spectrometer (UPLC-MS/MS, Agilent Technologies, Inc.). The details of the analytical method are described in Text S3. The different molecular masses of BDOM inside and outside the bags were named according to both the pyrolysis temperature and the molecular weight. For example, the BDOM sample pyrolyzed at 300 °C and isolated by 0.3 kDa dialysis bag was marked as “BD3-0.3I” (inside) and “BD3-0.3O” (outside).

2.5. Fluorescence quenching titration experiment

In order to explore the influence of pyrolysis temperature and molecular weight on the binding behavior between BDOM and the pharmaceuticals, BDOM pyrolyzed at 300 °C and 500 °C which were further isolated by 0.3, 1, 3, and 10 kDa cut-off molecular weight of dialysis bags (using the method in dialysis equilibrium experiment) were prepared. Based on the physicochemical properties of BDOM, the differences between 300 °C and 400 °C and 500 °C and >500 °C were small, so BDOM pyrolyzed at 400 °C and >500 °C were not considered in the fluorescence titration experiment. Moreover, CAP and CMZ having relatively higher hydrophobicity and binding capacity were selected as the representatives of pharmaceuticals (Yan et al., 2024). The fluorescence quenching titration experiment design is also shown in Fig. 1. Firstly, the fractionated BDOM (13 mL) inside and outside of the dialysis bags were introduced and diluted to approximately 1–2 mg L⁻¹ to maintain the same level of DOC concentration among different quenching systems. Then, CAP/CMZ titrations were added with the concentrations varying from 0.1 to 10 mg L⁻¹. It had been reported that the equilibrium time and binding capacity of BDOM-drug binding were correlated positively with the concentration of the drug (Yang et al., 2020a). Therefore, a relatively high concentration of the drugs was still selected in this study to better understand the binding capacity between BDOM and the drugs despite the low concentration of drugs in the environment. Each group was set up in three parallels and shaken for 48 h at 25 °C in the dark to ensure binding equilibrium (Huang et al., 2022). BDOM without drugs was used as the control. All BDOM solutions were subjected to UV-vis

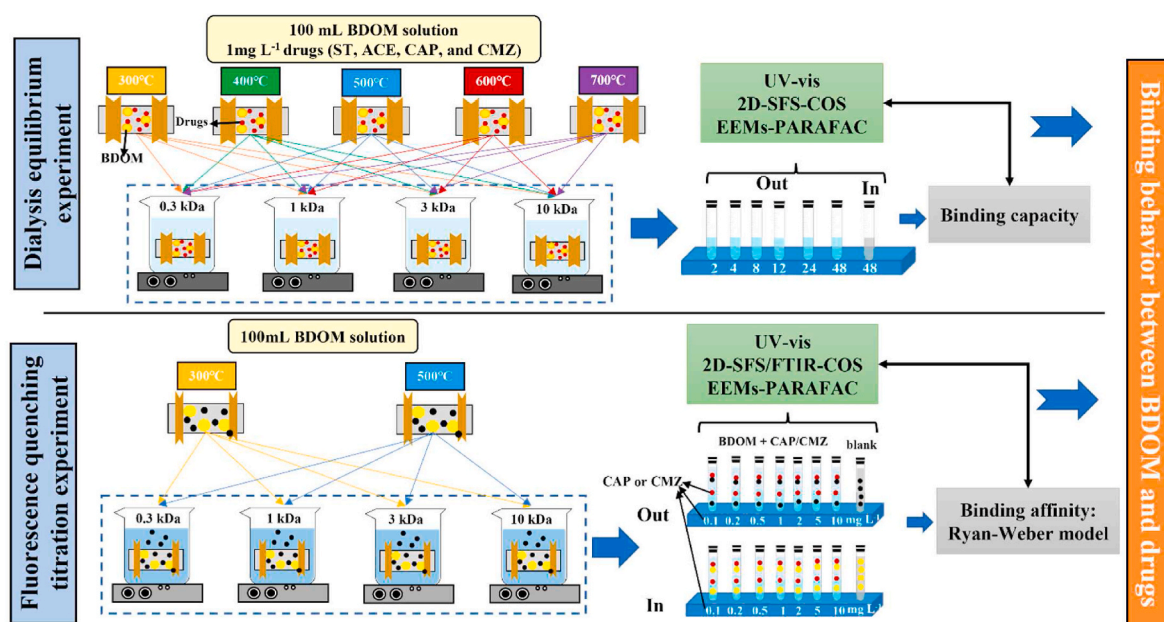


Fig. 1. The schematic diagram of the experiment of the pharmaceutical vs. BDOM binding behavior.

absorption spectra, SFS, and 3D-EEM spectroscopy, which were all ultrasonically dispersed before performing the measurements.

2.6. Characterization of BDOM

Multi-technologies proceeded in this study to characterize the physicochemical properties of BDOM, including scanning electron microscopy (SEM, S-3400N, Hitachi, Japan), UV-vis absorption spectrometer (UV3300, Mapada, Shanghai), Fourier transform infrared spectroscopy (FTIR, Spectrum One, PerkinElmer, USA), and fluorescence spectrophotometer (F-7100, Hitachi, Japan). The details of the description for these detections could be obtained from Text S4.

2.7. Data analysis

2.7.1. UV-vis parameters

Owing to the weakness of the traditional method of measuring the dissolved organic carbon (DOC) concentration such as time-consuming and requires a relatively larger sample, a quicker method using UV-vis absorption spectroscopy was proceeded in this study to estimate the DOC concentration according to the previous study (Liu et al., 2019). This method considered both the colored dissolved organic matter concentration and the lower/high molecular weight and aromaticity fractions. It was calculated by Equations (1) and (2):

$$f = \frac{1.135 \times \frac{UV_{254}}{UV_{365}} - 2.813}{\frac{UV_{254}}{UV_{365}} - 1.797} \quad (1)$$

$$DOC = \frac{A_{254}/r}{0.0232f + 0.0642(1-f)} \quad (2)$$

where A_{254} is the absorbance at 254 nm, r is the light path (cm), and UV_{254}/UV_{365} is the ratio of absorbance at wavelength 254 nm and 365 nm; if the f value < 0 or > 1 in the calculation, it will be assumed as 0 or 1, respectively.

UV_{253}/UV_{220} was defined as the ratio of absorbance at 253 nm–220 nm, which was usually employed to indicate the aromatic ring replacement (Fuentes et al., 2006). The low UV_{253}/UV_{220} value means that the aromatic ring was dominated by aliphatic chains, conversely, the aromatic ring was predominated by carboxyl and carbonyl groups (Fuentes et al., 2006).

2.7.2. PARAFAC modeling

In this study, the PARAFAC model was used to identify the fluorescent components based on the 3D-EEM data from the dialysis experiment (including the non-drugs BDOM samples, $n = 290$ samples in total) and the quenching titration experiment (300 °C quenching system: $n = 256$; 500 °C quenching system: $n = 256$). Before modeling, the primary Rayleigh and Raman scatter were removed and all data were corrected using the Raman peak of Milli-Q water. The detailed information has been reported in our previous study (Yan et al., 2023). Then, the substance determination of each fluorophore analyzed was compared with the online OpenFluor database platform (<http://www.OpenFluor.org>) (Murphy et al., 2014). Consistency of composition between PARAFACs from different studies was ensured by the Tucker consistency coefficient.

2.7.3. 2D-COS

2D-COS could uncover the sequence of the influence of external factors on different components of DOM (Noda, 2006). In this study, 2D-SFS-COS was used to analyze the diffusion order of the fluorescent component during the dialysis equilibrium and to investigate the quenching sequence of the fluorescent component by the pharmaceuticals using 2D Shige software (Kansai Prefectural University, Japan). 2D-FTIR-COS was used to describe the reaction process of functional groups to drugs. Moving window 2D-COS (MV-2DCOS) combining SFS was devoted to recognizing the transition points of spectral variations

along the perturbation variable direction (i.e., the drug concentration). Heterospectral spectra 2D-COS generated by SFS and FTIR were performed to analyze the relationships between fluorescent substances and functional groups. In these experiments, the equilibrium time and CAP/CMZ concentration were used as external interfering factors. A detailed explanation of the theory of 2D-COS analysis has been presented in previous studies (Fan et al., 2022; Fan et al., 2023; Noda, 1990).

2.7.4. Nonlinear complex model

The binding model was fitted to the binding process according to the study reported by Ryan and Weber (Ryan and Weber, 1982), so as to determine the binding parameters of the pharmaceuticals to the fluorescent components. The model assumed that the drug was bound to the equivalent and independent binding ligands with the stoichiometric ratio of the drug to ligand was 1:1. The binding parameters were calculated by Equation (3).

$$I = I_0 + (I_{ML} - I_0) \left(\frac{1}{2K_M C_L} \right) \left(1 + K_M C_L + K_M C_M - \sqrt{(1 + K_M C_L + K_M C_M)^2 - 4K_M^2 C_L C_M} \right) \quad (3)$$

where I_0 is the fluorescence intensity with no drugs; I represents the fluorescence intensity of BDOM at the pharmaceutical concentration of C_M ; I_{ML} is the fluorescence intensity limit when the fluorescence intensity no longer changes with increasing pharmaceutical concentration; K_M is the conditional stability constant ($\log K_M$); C_L is the total concentration of ligands in BDOM.

3. Results and discussion

3.1. Characterization of BDOM

The different apparent morphological characteristics of the frozen dried BDOM prepared at different pyrolysis temperatures images were shown in Fig. S1, as well as the biochar removed DOM or not. After the water extracted, small particles of BDOM were released from the biochar to some degree with a smoother surface for the DOM-released biochar. Moreover, the SEM images of BDOM and biochar are dramatically different. In general, the BDOM exhibited a type of polymer with smooth and irregular surfaces as well as disordered structure, interspersed and overlapped with each other. For the BDOM pyrolyzed at the lower temperature, it had an amorphous and irregular structure with discrete particle forms; while for the BDOM pyrolyzed at the higher temperature, the irregular structure progressively fragmented and its surfaces were smooth and rigid with fibrous structures.

The FTIR spectra of BDOM showed in Fig. S2 that the variations of the functional groups were concentrated in the regions from 3680 to 2780 cm^{-1} and 1800 to 500 cm^{-1} . The different assignments of the FTIR band corresponding to the functional groups are shown in Table S2. With increasing pyrolysis temperature, the intensity of phenolic or alcohol hydroxyl $-\text{OH}$ (3438 cm^{-1}), carboxyl $\text{C}=\text{O}$ (1700/1330 cm^{-1}), phenolic $-\text{OH}$ (1402 cm^{-1}), aliphatic $\text{C}-\text{OH}$ (1122 cm^{-1}), polysaccharides $\text{C}-\text{O}$ (1010 cm^{-1}), and carboxyl $-\text{OH}$ (621 cm^{-1}), showed an increased or decreased trend. For example, the intensity of peaks at 703 cm^{-1} and 832 cm^{-1} increased indicating a larger degree of condensation and the protein-like substance of BDOM increased as the pyrolysis temperature increased (He et al., 2021a; Özçimen and Ersoy-Meriçboyu, 2010). Whereas, the intensity of phenolic $-\text{OH}$ and carboxyl $-\text{OH}$ decreased or even disappeared with increasing pyrolysis temperature. When increasing the pyrolysis temperature, dehydration brought the loss of $-\text{OH}$ groups on biochar and the simultaneous release of water (Cheng et al., 2021). Similarly, this process would lead to the loss of $-\text{OH}$ on BDOM.

The UV-vis spectra, DOC concentration, and UV_{253}/UV_{220} of BDOM

pyrolyzed at 300–700 °C were displayed in Fig. S3. The absorbance of BDOM decreased exponentially when the wavelength was longer than 300 nm and was preponderant high at lower pyrolysis temperatures. The calculated DOC concentration of BDOM ranged from 1.09 to 147.65 mg L⁻¹ derived from different pyrolysis temperatures. It gradually decreased with the pyrolysis temperature increased from 300 °C to 400 °C, and then dramatically dropped down when the pyrolysis temperature was at 500 °C, after that, it slightly increased. These results were confirmed by many previous works (Gui et al., 2020; Liu et al., 2023). At lower pyrolysis temperatures (<500 °C), the feedstock experienced strong carbonization and decomposition. But at the higher pyrolysis temperature (>500 °C), the products might undergo rearrangement reactions such as decomposition, cyclization, polymerization, and condensation, thus resulting in a slight increase in DOC concentration (Liu et al., 2015). A similar trend was found for the UV₂₅₃/UV₂₂₀ value, indicating that the carboxyl groups on the BDOM pyrolyzed at lower temperatures were gradually replaced by the aliphatic chains when the pyrolysis temperature increased.

BDOM derived at different pyrolysis temperatures had varying fluorescence components. Based on the PARAFAC model, two humic-like components (C1 and C2) and one protein-like component (C3) were obtained and further compared with the models in the OpenFluor database (Fig. 2 and Table S3). C1 was associated with fluorescence peaks A and C, which was a humic-like substance that peaked at Ex/Em of 312/390 nm (Coble, 1996; Gullian-Klanian et al., 2021). The peak of C2 at Ex/Em of 360/445 nm was similar to the fluorescence characteristics of humic-like fraction, which was widely distributed in freshwater systems and was related to high molecular weight aromatic substances (Lambert et al., 2017). C3 peaked at Ex/Em of 280/334 nm, which was associated with protein-like fluorophores found in seawater, freshwater, and artificial environments, derived from biological activity (Catalán et al., 2021). Overall, the humic-like substance (C1) was the dominant component for all the BDOM regardless of the pyrolysis temperature. The fluorescent intensity of the BDOM produced at 300 °C and 400 °C was much higher than that of BDOM produced at the temperature higher than 500 °C. With the pyrolysis temperature increased, the percentage of

the protein-like substance increased from 15.33% (300 °C) to 20.46% (500 °C), which was in accordance with many previous studies (He et al., 2021a; Yan et al., 2023).

3.2. Binding capacity of the pharmaceuticals to BDOM

3.2.1. Changes in the spectral properties for the different molecular weights of BDOM

After dialysis separation, the smaller molecular weight of BDOM inside the bag was transferred outside the bags with DOC concentration and UV₂₅₃/UV₂₂₀ increased significantly for BDOM outside the bags (Fig. S4). Apparently, the DOC concentrations of BDOM derived from the lower pyrolysis temperatures (i.e., 300 °C and 400 °C) were much higher than those from the higher pyrolysis temperatures (i.e., >500 °C). It was agreed with the literature that the DOC concentration in BDOM derived from different pyrolysis temperatures with the same molecular weight (i.e., <1 kDa) decreased with the pyrolysis temperature increased, which meant that the BDOM derived from the lower pyrolysis temperature possessed a higher DOC concentration (Yu et al., 2023). Besides, a gradually increased DOC concentration for the larger molecular weight of BDOM was observed with the pyrolysis temperature increase. These results revealed that more BDOM with oxygen-containing groups on aromatic rings diffused to the outside of the bag for the larger molecular weight of BDOM when the pyrolysis temperature increased. This is possibly ascribable to the higher pyrolysis temperature has a strong cleavage effect on BDOM, which results in a smaller molecular weight of BDOM (Li et al., 2022; Zhang et al., 2020b). Therefore, with the cut-off molecular weights of the dialysis bag increased, the smaller molecular weight of BDOM pyrolyzed at a higher temperature will easily diffuse outside of the solution, which contributes to the high DOC concentrations for the large molecular weight of BDOM (e.g., <10 kDa) derived from the higher pyrolysis temperature.

In order to visualize the sequence of the fluorescent components diffused during the dialysis process, the synchronous and asynchronous maps of 2D-SFS-COS were plotted (Fig. S5). In the synchronous maps (Table S4), an auto-peak at 280 nm representing the tryptophan-like

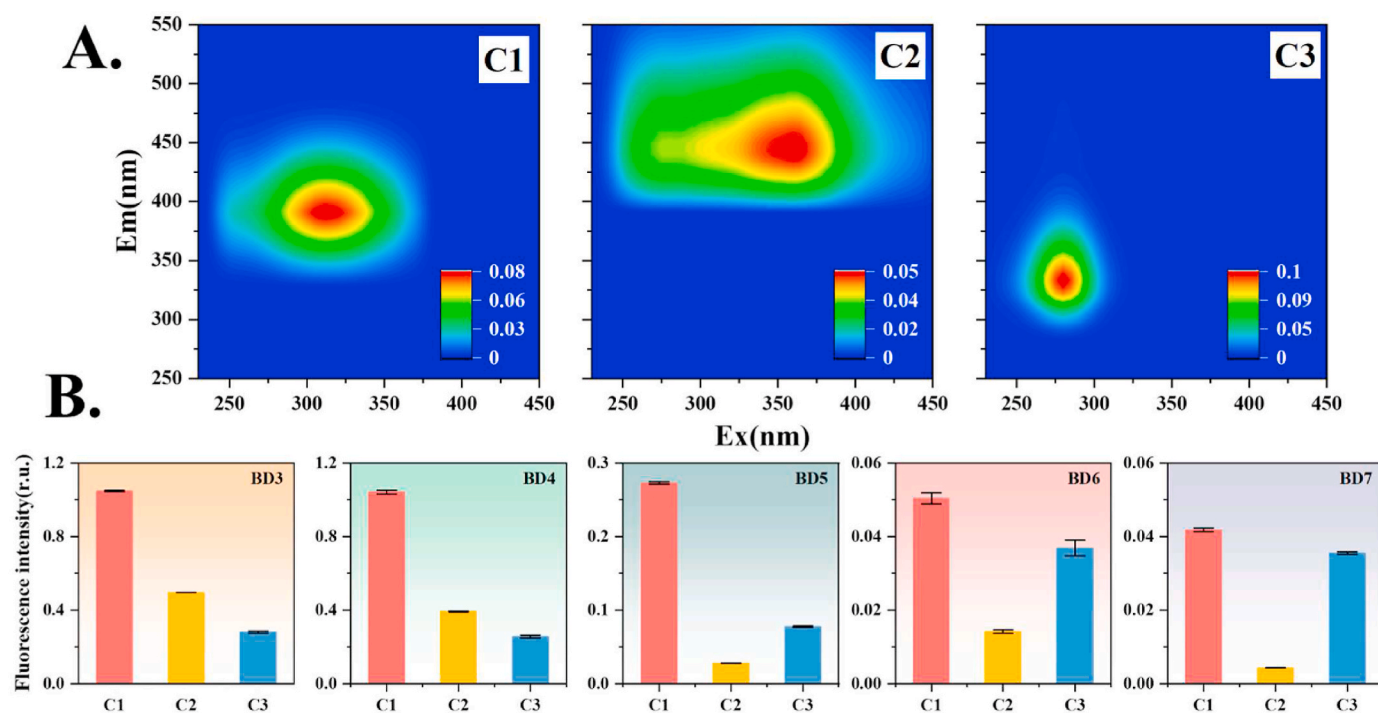


Fig. 2. 3D-EEMs of the identified PARAFAC components (A.) and the fluorescence intensity of the PARAFAC components for the BDOM pyrolyzed at 300–700 °C (B.).

substance was shown in all molecular weights of BDOM derived at the pyrolysis temperature higher than 400 °C. Besides, another auto-peak displayed around 330 nm representing the fulvic-like substance was found in all molecular weights of BDOM derived from the lower pyrolysis temperatures (i.e., 300 °C and 400 °C). These results suggested that the tryptophan-like material in BDOM derived from the relatively higher pyrolysis temperatures was more sensitive during dialysis, yet only the fulvic-like material in BDOM produced at the lower pyrolysis temperatures separated sensitivity through dialysis. By combining the signs (positive and negative) of the cross-peaks (bottom-right) in both synchronous and asynchronous maps and Noda's principle, the diffused sequence of the fluorescence components could be gained (Table S5). Except for all the molecular weights of the BD5 systems and BD6-0.3 system, the sequence of fluorophore diffusion in BDOM was generally ordered as 210 > 280 > 330 > 265/350 nm, demonstrating that the protein-like material diffused outside the dialysis bag before the humic-like material. Consequently, it could be concluded that the protein-like substance was more susceptible and released preceding than the humic-like substance during the dialysis equilibrium process, which would further impact the diffusion behavior of pharmaceuticals.

Furthermore, the fluorescence components went through to the outside of the bags and the fluorescent intensity of the BDOM outside the bags gradually increased after the dialysis process. As shown in Fig. S6, C3 was the dominant component for all the small molecular weights of BDOM outside the bags. As the pyrolysis temperature increased, the average percentage of C3 in BDOM outside the four cut-off molecular weights bags significantly increased from 48.86% at a pyrolysis temperature of 300 °C to 88.40% at a pyrolysis temperature of 700 °C after reaching the equilibrium; while the counterpart for C1 and C2 showed the contrary tendencies. After being separated by the dialysis bags, the average percentage of C3 decreased from 58.8% to 54.34% at the beginning to 48.86% and 53.18% at the equilibrium time for BD3 and BD4 systems outside the bags; while the corresponding values increased from 61.96%, 76.12%, and 57.74%–88.30%, 94.48%, and 88.40% for BD5, BD6, and BD7. Compared to BDOM outside the bags, a different percentage distribution of the fluorescent component was represented

for the BDOM inside the bags. Namely, a lower percentage of C3 was related to the BDOM derived at the lower pyrolysis temperature (300 °C: 20.56%; 400 °C: 34.36%), whereas it dramatically increased to a higher percentage level for the BDOM pyrolyzed at the temperature higher than 500 °C (500 °C: 82.36%; 600 °C: 85.60%; 700 °C: 55.98%). However, no distinct contrast was observed for the distribution of the percentage of the fluorescent components between the dialysis bags with different cut-off molecular weights. Given the above, it illustrated that: 1) the protein-like substance was the dominant fluorescent component for the small molecular weight of BDOM outside the bags separated by the dialysis bags, in particular for BDOM pyrolyzed at higher temperatures; 2) the humic- and the protein-like substance was the dominant component for the large molecular weight of BDOM inside the bags pyrolyzed at lower and the higher temperature, respectively; 3) the extend that the protein-like substance diffused more than those for the humic-like substance with dialysis process increased as the pyrolysis temperature increased. A similar profile in the protein-like materials abundant in the small molecular weight of particles had been observed in the natural DOM (Romera-Castillo et al., 2014; Yan et al., 2018).

3.2.2. Binding capacity of the different molecular weights of BDOM with the pharmaceuticals

In general, the concentration of the pharmaceuticals increased rapidly within 8 h and gradually reached stable after 24 h (Fig. S7). After checking the equilibration across the membrane (48 h), the concentrations of the pharmaceuticals in and outside the bag were both determined (C_{in} and C_{out}). If $C_{bind} = C_{in} - C_{out} > 0$, BDOM-drugs binding occurs. The higher the C_{bind} , the stronger the binding capacity. As shown in Fig. 3A, the average value of C_{bind} for the four molecular weights dialysis systems was 0.98, 2.23, 3.48, and 11.28 $\mu\text{g L}^{-1}$ for ST, ACE, CAP, and CMZ, which indicated that the stronger and weaker binding capacity was related to the BDOM-CMZ/CAP and BDOM-ST/ACE system. It was confirmed by the higher positive relationship between the concentration of ST and ACE ($R^2 = 0.98$) and CAP and CMZ ($R^2 = 0.74$) outside the bags during the dialysis process (Fig. S8-Total), which revealed the similar binding behavior between them. Notably, this trend was

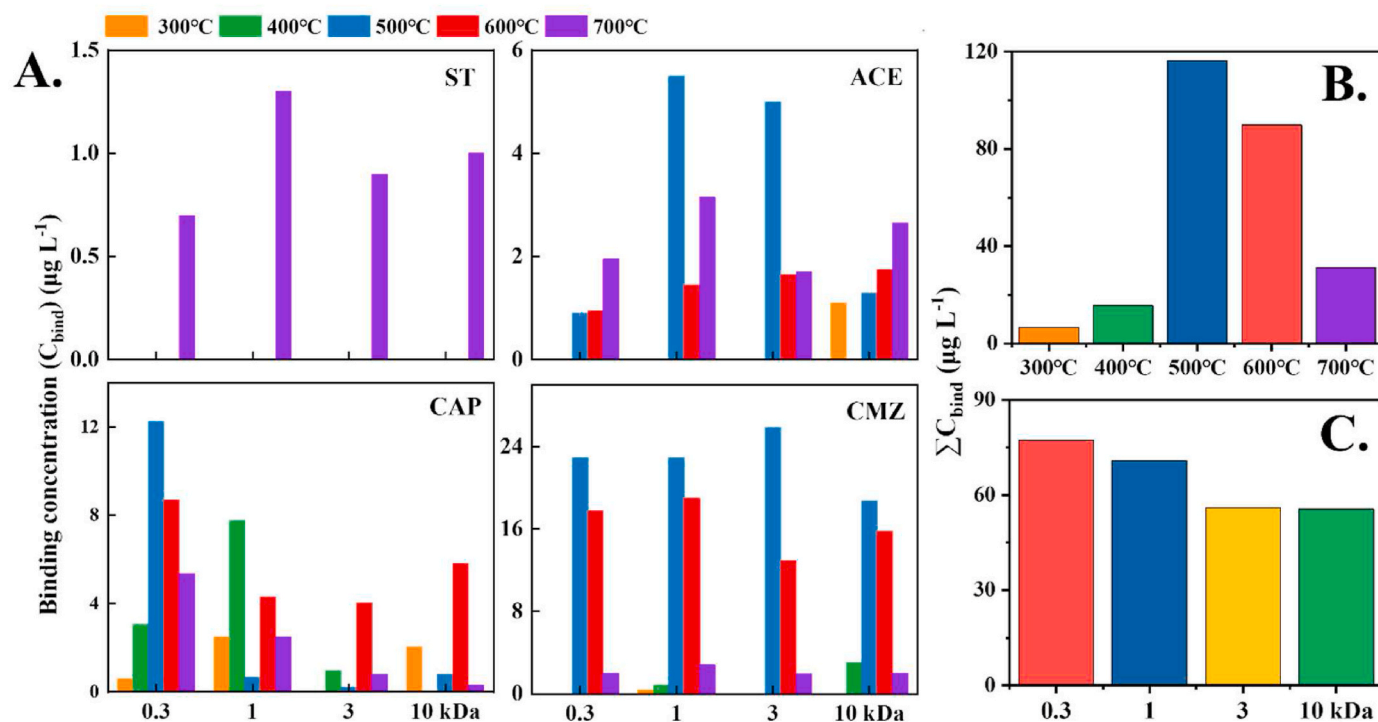


Fig. 3. The concentration of the pharmaceuticals binding on BDOM (A.) and the sum of the binding concentration for all pharmaceuticals at different pyrolysis temperatures (B.) and in different molecular weight systems (C.).

precisely related to the hydrophobicity of the drugs ($\log K_{ow}$) (Table S1), with $R^2 = 0.95$. The previous study found that hydrophobicity was an important factor controlling the interaction between the natural aquatic DOM and the pharmaceuticals (Yan et al., 2015). That is, the hydrophobicity deeply impacted the binding capacity between the BDOM and the pharmaceuticals. In addition, most of the C_{bind} values were negative for the BDOM derived from the lower pyrolysis temperatures (i.e., 300 °C and 400 °C) (Fig. 3A). Besides, the C_{bind} values of BDOM derived from the pyrolysis temperature of 500 °C were high (Fig. 3B). It meant that the binding capacity between the pharmaceuticals and BDOM strengthened with the pyrolysis temperature increased which had the peaking value at 500 °C, but it turned to be weak when the pyrolysis temperature was higher than 500 °C. Combined with the above results in FTIR, it could be known that less binding behavior occurred between the pharmaceuticals and BDOM pyrolyzed higher than 500 °C might be attributed to the decomposition of BDOM reducing the binding sites at the higher pyrolysis temperature. Noteworthy, the binding behavior between BDOM and the drugs relied on the pyrolysis temperature. That is, a similar binding behavior was observed among all drugs and the BDOM pyrolyzed at 300 °C and 400 °C with an R^2 value higher than 0.97, whereas a totally different binding behavior was found for ACE and the BDOM pyrolyzed at temperatures higher than 500 °C with R^2 value lower than 0 (Fig. S8). Besides, the C_{bind} values of BDOM derived from the molecular weight of 0.3 kDa systems were relatively high (Fig. 3C). It was due to the small molecular weight dialysis system that could capture more BDOM which could bind more pharmaceuticals.

3.2.3. Impact of the spectral properties on the binding capacity of BDOM-pharmaceuticals

In order to explore the impact of the spectral properties on the binding capacity of the pharmaceuticals to BDOM, intercorrelation analysis and redundancy analysis (RDA) were conducted using the

corresponding data for outside the bags. For all data in dialysis systems (Fig. 4A-Total), the explanation for the first and second axes was 86.05% and 13.83%, respectively, suggesting that the binding capacity between BDOM and the pharmaceuticals was significantly influenced by spectral characteristics. Moreover, C3 had higher eigenvalues of PC1, while the other parameters had higher eigenvalues of PC2. Besides, the concentration of CAP and CMZ was positively correlated to C3 with high values of PC1, while ST and ACE had small angles and high eigenvalues of PC2. It indicated that the protein-like substance greatly impacted on the binding capacity of BDOM-CAP&CMZ, while the humic-like substance having high DOC concentration and more carboxyl and carbonyl groups on aromatic rings played a vital role in the binding capacity of BDOM-ST&ACE.

Although the difference in the eigenvalues and explanations between the dialysis systems with different molecular weights was small, an increasing trend was shown for the explanation of PC1 with the cut-off molecular weight of the dialysis bag increasing (Fig. 4A 0.3–10 kDa), which illustrated that the larger molecular weight of BDOM with more protein-like substances had more impact on binding with CAP and CMZ. In addition, a completely different RDA result was observed between the pyrolysis temperature lower and higher than 500 °C (Fig. 4A 300–700 °C). For BD3 and BD4 systems, all spectral parameters and pharmaceuticals had high eigenvalues of PC1, suggesting that the spectral properties seemed to influence the binding behavior of the pharmaceutical jet without any specific spectral feature related to the binding process. This phenomenon might be owing to less binding behavior occurring between them (Fig. 3), which was confirmed by the dramatically decreased eigenvalues for ST and ACE (nearly 0) when the pyrolysis temperature was higher than 500 °C. In other words, less impact was impacted by the spectral properties of BDOM on the binding behavior of ST and ACE; while for CAP and CMZ, the spectral parameters of BD5-BD7 were affecting the binding process with C3 keeping positive

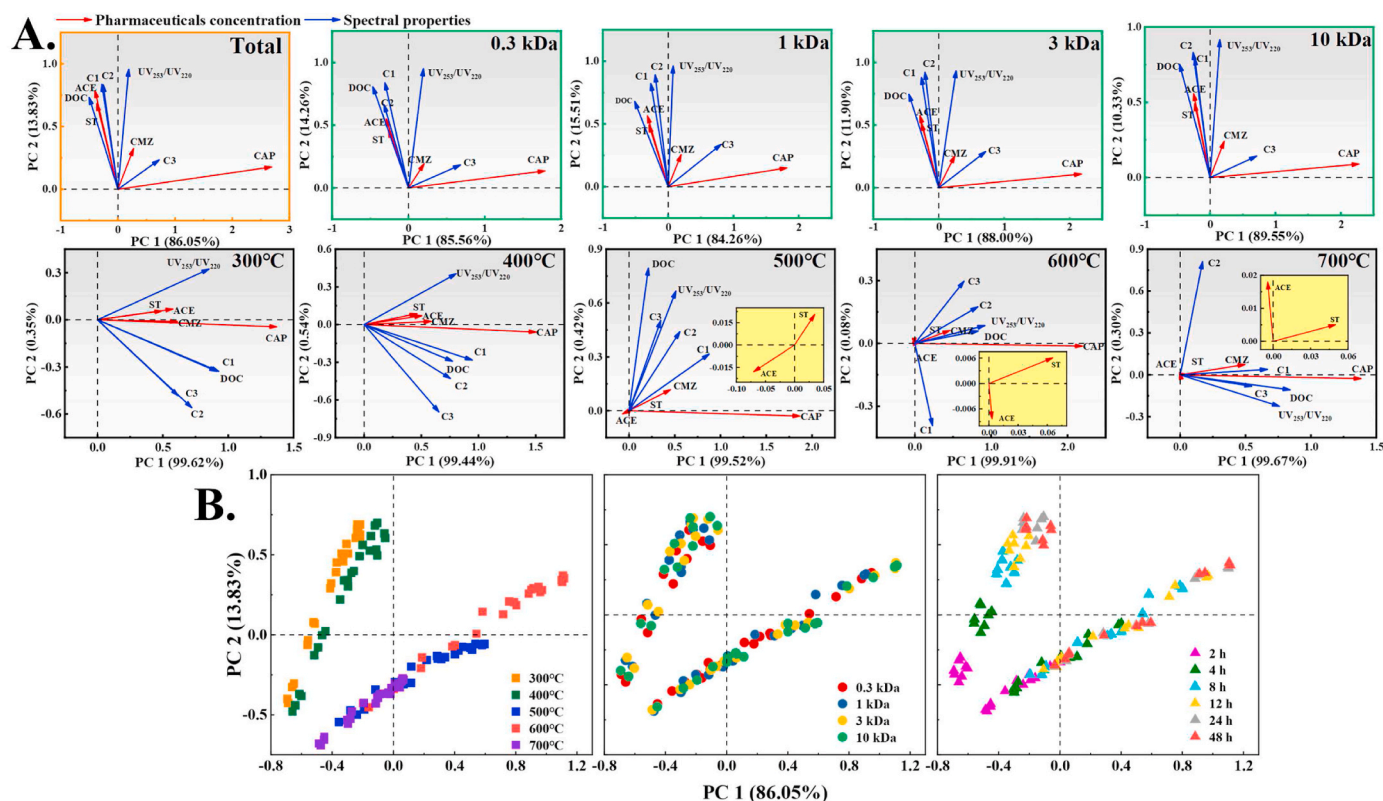


Fig. 4. RDA results (A.) and scores of samples (B.) between the pharmaceuticals and the spectral properties of BDOM during the dialysis equilibrium process (orange frame: all data; green frame: the data of the dialysis process for the different molecular weights; black frame: the data of the BDOM derived from different pyrolysis temperatures). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

correlation to CAP and CMZ.

Moreover, the data of the BDOM from different pyrolysis temperatures and sampling times yield different groups along the RDA axis, whereas a similar trend was not displayed for the data of the BDOM from different cut-off molecular weight bags (Fig. 4B). In general, the data of the pyrolysis temperature lower (BD3 and BD4) and higher (BD5-BD7) than 500 °C was located at the left- ($PC1 < 0$) and right-side ($PC1 > 0$) of the score plot, illustrating that the protein-like substance highly associated with the binding behavior between BDOM pyrolyzed at higher temperatures and the pharmaceuticals. Besides, the more the dialysis time, the higher the scores of both PC1 and PC2, indicating the more influence of the spectral properties on the binding of pharmaceuticals with increasing dialysis time.

3.3. Binding mechanism of BDOM and pharmaceutical

3.3.1. Binding characteristics of BDOM with pharmaceuticals by SFS

In this study, SFS was initiated to qualitatively describe the ability of the fluorescent components of BDOM to bind with CAP and CMZ (Biet al., 2016). As Fig. S9 shows, the intensity of the fluorescence peak near 280 nm (tryptophan-like materials) and 345 nm (fulvic-like materials) in BDOM decreased to various degrees with the concentrations of CAP and CMZ increased. Overall, the fluorescent intensity of the

tryptophan-like fluorophores declined more than that of the fulvic-like substance, particularly for the BD5 systems, indicating that the protein-like substance was more likely to bind with CAP and CMZ. In the BD3-CMZ systems, the fluorescent intensity of the tryptophan-like fluorophores decreased more than that of BD3-CAP systems, revealing the stronger binding ability of the tryptophan-like substance in BD3 with CMZ. Additionally, the more declined fluorescent intensity of the tryptophan-like fluorophores was observed for the small molecular weights of BD3 (outside the bag). This might be explained by the transfer of the protein-like substances to the outside of the bags during the dialysis separation. In contrast, a stronger binding ability was found between the tryptophan-like fluorophores in BD5 and CAP, whereas the small molecular weights of BD5 (outside the bag) were more highly bound with CMZ. As the concentration of CMZ increased, the fluorescence peak at 285 nm in BD5 was red-shifted from 285 nm to 305 nm, and the number of red-shifted wavelengths correlated with the quenching effect. It indicated that the conformation of the tryptophan-like component of BDOM changed when it interacted with CMZ (Bani-Yaseen, 2011).

3.3.2. Binding disturbing sites and sequence of BDOM with pharmaceuticals by 2D-SFS-COS

The MV-2DCOS analysis combining SFS was devoted to observing the

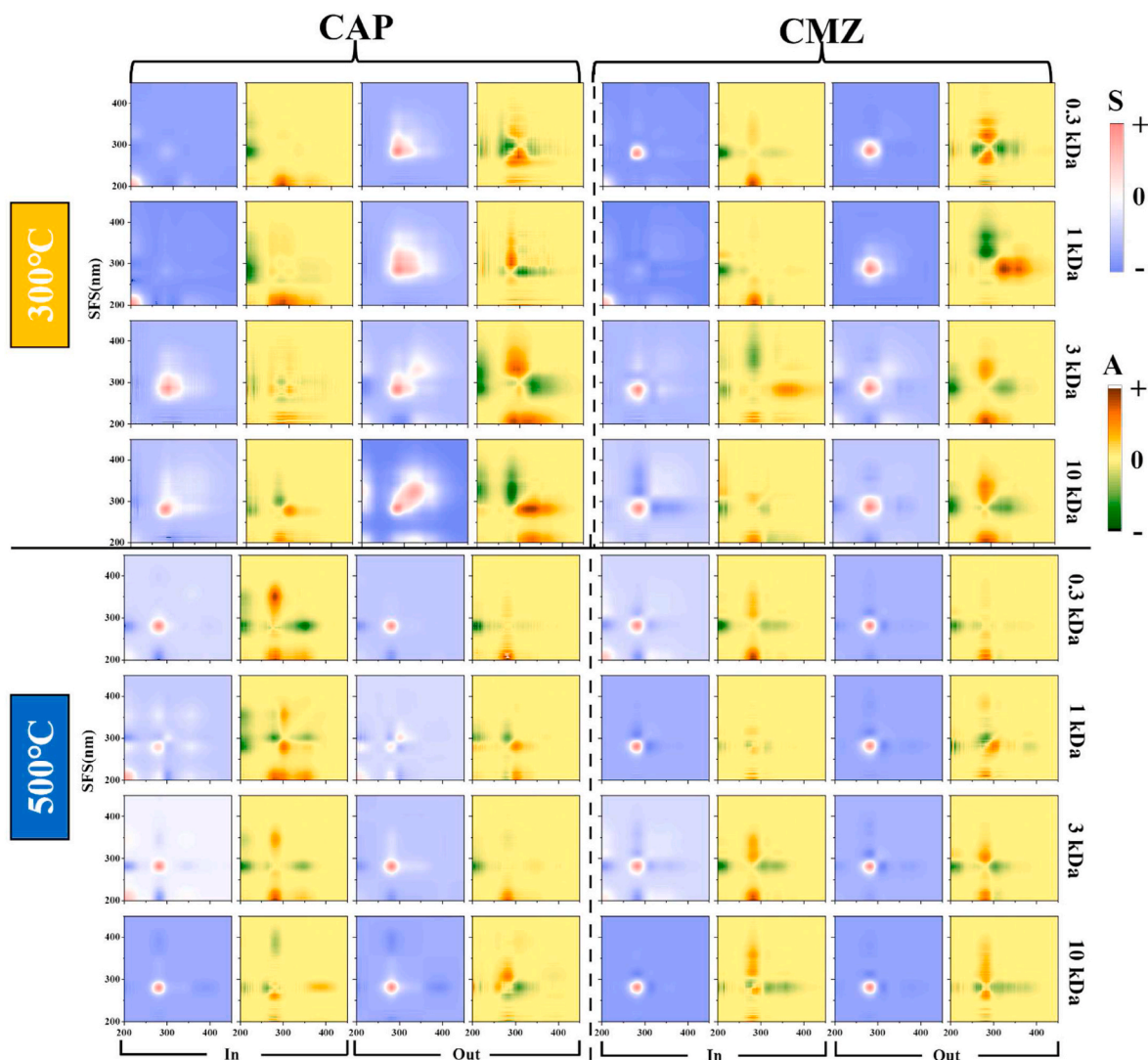


Fig. 5. Synchronous (S) and asynchronous (A) of 2D-COS maps obtained from the SFS of BDOM derived from biochar pyrolyzed at 300 °C and 500 °C with CAP/CMZ titrated.

specific region of significant spectral intensity change with the drug concentration increasing, which had been used in revealing the eigenvalues of copper concentration bounding with the different fluorescent components (Liu et al., 2022; Wu et al., 2022). The results in Fig. S10 showed that the main cross peaks appeared around 280 nm (tryptophan-like) when the drug concentration was 2–3 mg L⁻¹ and 8–9 mg L⁻¹, especially for BDOM pyrolyzed at 500 °C. This phenomenon implied that the protein-like substance in BDOM could react with the drugs at different drug concentrations to trigger transitions of the conformation. Additionally, more cross-peaks were displayed for BDOM pyrolyzed at 300 °C (e.g., BD3-30 vs. CMZ and BD3-100 vs. CAP), indicating more fluorophores in BD3 were bound to the different concentrations of pharmaceuticals.

To reveal the order and sensitivity of the fluorescence components in BDOM to bind with CAP/CMZ, the 2D-SFS-COS technique was used. The synchronous and asynchronous correlation maps obtained are shown in Fig. 5. There were many positive auto-peaks along the diagonal of the synchronous map around 210, 285, 330, and 350 nm corresponding to the tyrosine-, tryptophan-, fulvic-, and humic-like components, suggesting that these fluorophores might have had different degrees of binding to CAP and CMZ. Wherein, the highest fluorescence intensity at 285 nm was characterized, indicating that the tryptophan-like substance was more susceptible to the addition of CAP and CMZ. These results were in accordance with many previous studies (Huang et al., 2022; Wang et al., 2016; Yuan et al., 2019). In general, more fluorophores were binding with CAP (including both the protein- and

humic-like substance) compared to CMZ (primarily the protein-like substance). Nonetheless, no apparent difference in the binding behavior referring to the sensitivity was shown between the pharmaceuticals and the BDOM with different molecular weights and pyrolysis temperatures. Furthermore, some negative cross-peaks were discovered at 280–300/210 and 285–288/315–360 on the off-diagonal lines of the synchronous map for most of the binding systems, suggesting that the tryptophan-like substance was reduced proceeding in the opposite direction to the tyrosine- and the humic-like substance with the CAP/CMZ adding.

The sequence of BDOM fluorescence components bound with CAP and CMZ was determined based on the positive and negative signs (Tables S6–S8) of cross peaks in the synchronous and asynchronous maps with Noda's principle. Overall, the preferential binding of CAP with BDOM could be ordered as the tyrosine- > tryptophan- > the fulvic- or humic-like substance (210 > 280 > 300/330/350/388 nm); the counterpart for CMZ was ordered as the tyrosine-/(>) the humic- > tryptophan-like substance (210/315 > 280 nm or 210 > 315 > 280 nm). Likewise, no remarkable difference in the binding behavior with respect to the binding sequence was found between the pharmaceuticals and the BDOM with different molecular weights and pyrolysis temperatures. Combining the above results in 2D-COS, it could be concluded that the protein-like substance was more susceptible and preferentially bound with the pharmaceuticals, which was in line with sulfadiazine combined with soil DOM (Yuan et al., 2023).

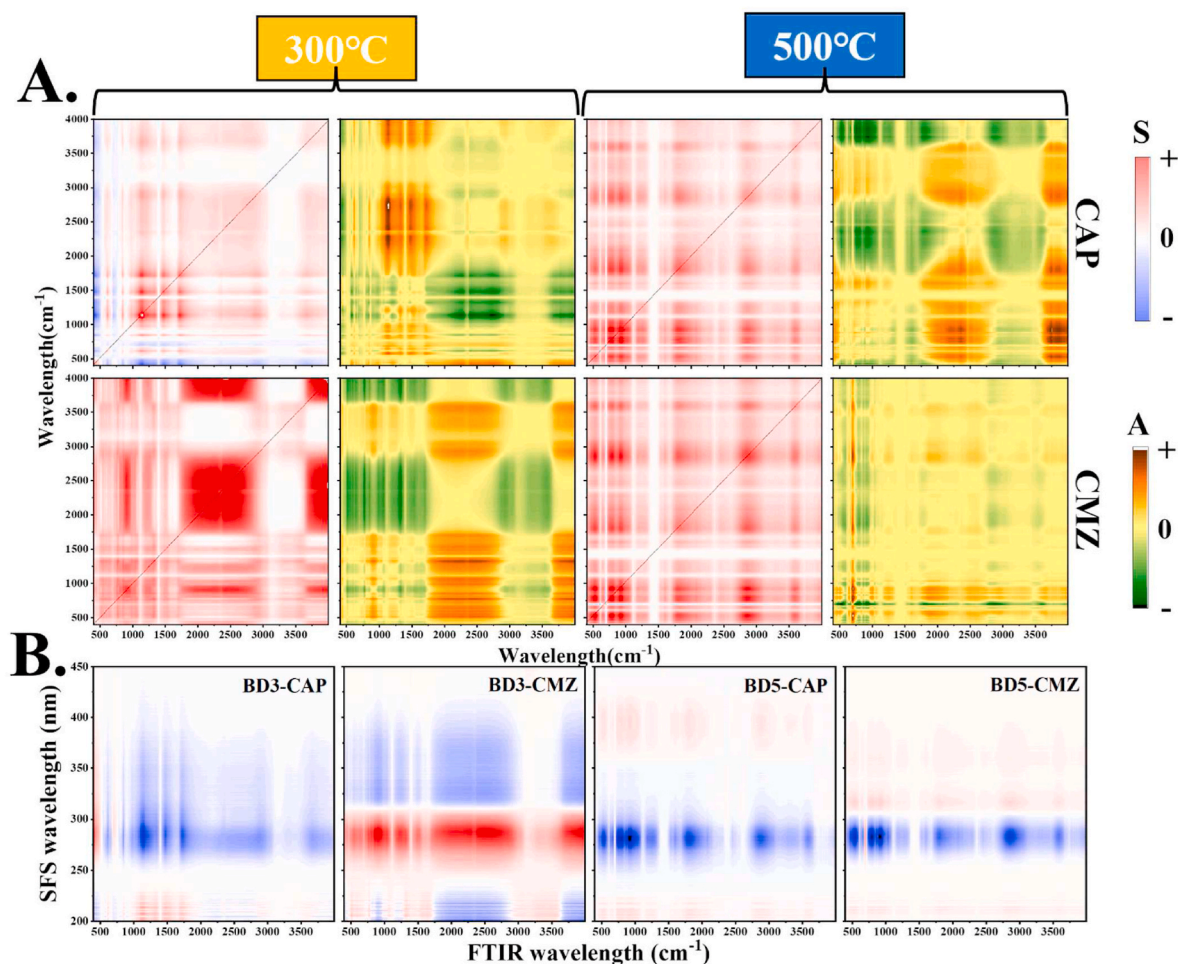


Fig. 6. Synchronous (S) and asynchronous (A) of 2D-COS maps obtained from the FTIR of BDOM derived from biochar pyrolyzed at 300 °C and 500 °C with CAP/CMZ titrated (A.). Synchronous map from the heterospectral 2D-COS generated from the SFS and FTIR of BDOM derived from biochar pyrolyzed at 300 °C and 500 °C with CAP/CMZ titrated (B.).

3.3.3. Binding sequence of BDOM with pharmaceuticals by 2D-FTIR-COS

In the synchronous maps of BDOM-CAP/CMZ systems, BD5 systems generally had more auto-peaks after the wavelength of 1800 cm^{-1} than those in BD3 systems (Fig. 6A), indicating more functional groups in BD5 were involved with the binding of the pharmaceuticals in these bands. Moreover, the auto-peaks in CMZ systems had higher intensities than those in the corresponding CAP systems, revealing stronger binding occurred between BDOM and CMZ. In detail, there are nine auto-peaks displayed at $580/502$ (A), $850/890/780$ (B), 920 (only in BD5) (C), $1132/1243/1280$ (D), $1480/1550$ (E), $1724/1798$ (F), 2487 (only in BD5) (G), $2821/2850$ (H), and $3655/3580\text{ cm}^{-1}$ (I), which assigned to the C–O–C bonding in aromatics groups (A), aromatic C–H (B), O–H deformation and C–C=O bend in carboxylic acids (C), aliphatic C–OH (D), amide II N–H (E), carboxyl C=O (F), nitrile group C≡N (G), aliphatic C–H (H), and phenolic-OH (I) (Table S2), suggesting the abundant oxygen functional groups in BDOM bound with the two drugs. Furthermore, the differences in the binding sites between BD3 and BD5 were more those between CAP and CMZ, illustrating that the pyrolyzed temperature impact is stronger on the binding behavior of pharmaceuticals than that of pharmaceutical types. Based on the assignments of the cross-peaks in the 2D-FTIR-COS maps (Tables S9 and S10), the binding sequence of the functional groups in BD3 to CAP was $E > D > F > B > I > A > H$; the BD3 binding to CMZ complied the sequence of $B > D > E > A > I$; the sequence of BD5 binding to CAP was $G > F > I > A > B > C > D > H > E$; and the binding sequence for BD5 to CMZ was $G > D > F > E > H > I > A > B > C$. Generally, the common oxygen or nitrogen-containing functional groups such as aliphatic C–OH, C=O in carboxyl, nitrile group C≡N, and amide II N–H possessed higher attractions and binding **preferentially** with the pharmaceuticals, as well as the aromatic C–H (Fig. 7). Because of the dominated ligand exchange and H-bonding of these functional groups, they had been shown to exhibit proton and metal binding abilities in previous studies (Fan et al., 2023; Huang et al., 2023).

To verify the correspondence between fluorescence and FTIR spectra, heterospectral 2D-COS was employed based on SFS and FTIR which represent different features of the BDOM structure (Xu et al., 2018). As displayed in Fig. 6B (the asynchronous map), the FTIR bands at $580/502$, $850/890/780$, 920 (only in BD5), $1132/1243/1280$,

$1480/1550$, $1724/1798$, 2487 (only in BD5), $2821/2850$, and $3655/3580\text{ cm}^{-1}$ were positively or negatively correlated with the fluorescent band at 210 , 285 , 310 (in BD5-CMZ), 350 (in BD3), and 380 (in BD5-CAP) nm. These results suggested that both the protein- and the humic-like substance were related to those functional groups binding with the drugs including aliphatic C–OH, amide II N–H, carboxyl C=O, and phenolic-OH, etc. A similar phenomenon was observed for the natural DOM derived from a shallow macrophytic lake (Fan et al., 2022). Besides, the tryptophan-like substance (SFS = 280 nm) showed a higher correlation with these functional groups. Combining the above results, it could be known that the binding behavior between BDOM and the pharmaceuticals mainly occurred on the aliphatic C–OH, amide II N–H, carboxyl C=O, and phenolic-OH in the tryptophan-like substance.

3.3.4. Binding affinity of BDOM with pharmaceuticals by 3D-EEM

To accurately depict the binding behavior of the independent fluorescent components with drugs while avoiding the overlap of fluorescence peaks in SFS, the PARAFAC model was introduced and four fluorescence components were both separately validated in the BD3 (C1–C4) and BD5 (C5–C8) systems (Fig. S11 and Table S11). In BD3 systems, three humic-like components (C1, C2, and C4) and one protein-like component (C3) were identified. Specifically, C1 peaked at Ex/Em = $320/388\text{ nm}$ and was assumed to be the humic-like component (Lin and Guo, 2020). C2 was similar to the humic-like component which was previously found in wastewater/nutrient enrichment tracers and produced by microbial degradation with a maximum peak at Ex/Em = $352/432\text{ nm}$ (Murphy et al., 2011). C3 showed a maximum peaking at Ex/Em = $285/398\text{ nm}$ which was the protein-like substance (Murphy et al., 2018). C4 had similar fluorescence properties to terrestrial humic-like substances, with a peak at Ex/Em = $270(383)/482\text{ nm}$, which was widely found in freshwater environments (Lambert et al., 2016). In contrast, three humic-like fractions (C5, C7, and C8) and one protein-like fraction (C6) were identified from BD5 systems. C5 was the humic-like component with a maximum peaking at Ex/Em = $310/408\text{ nm}$ (Catalá et al., 2015). The C6 peaked at Ex/Em = $280/366\text{ nm}$ corresponding to the protein-like fluorescence peak, which was further classified as the tryptophan-like component (Yamashita et al., 2011). C7 peaking at Ex/Em = $304/378\text{ nm}$, was similar to the humic-like

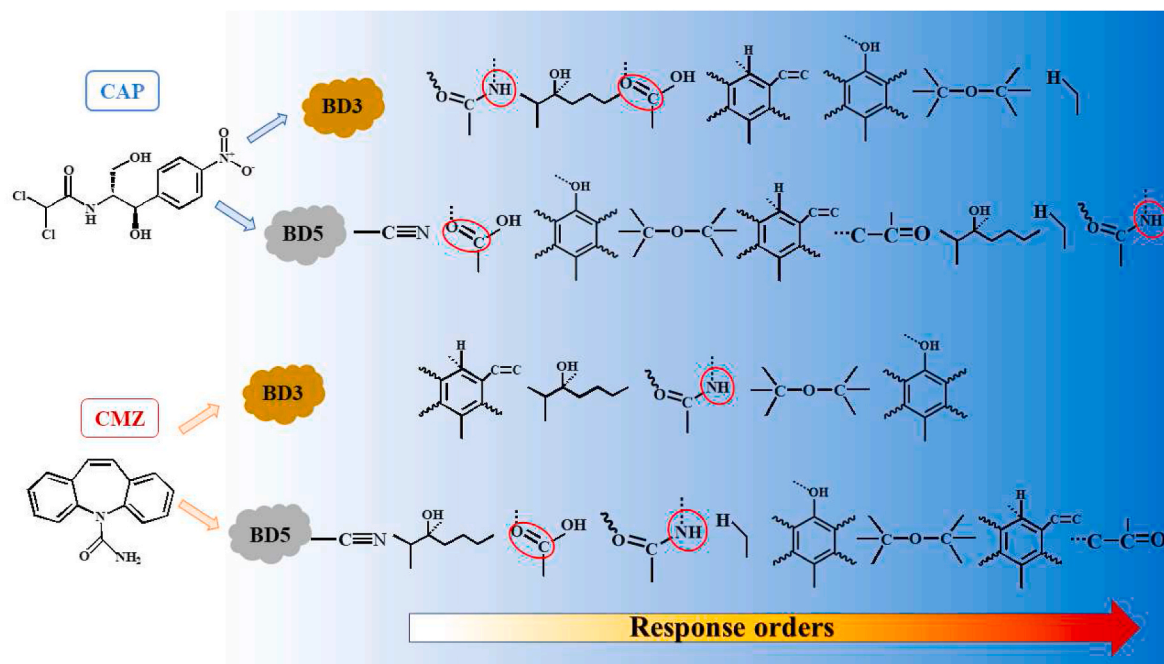


Fig. 7. Binding affinities of different functional groups in BDOM derived from biochar pyrolyzed at $300\text{ }^{\circ}\text{C}$ and $500\text{ }^{\circ}\text{C}$ with CAP/CMZ. The CAP/CMZ binding affinities decrease from left to right.

fluorophore (peak M) and was found in the ocean, asserted to be the humic-like component related to microbial activity (Hong et al., 2021). C8 was a terrestrial humic-like material with a peak at Ex/Em = 230 (370)/465 nm, which was widely observed in aquatic environments (Lin and Guo, 2020).

The fluorescence fractions of BDOM pyrolyzed at different temperatures and isolated from different cut-off molecular weights of bags exhibited distinct response characteristics after the addition of different concentrations of CAP and CMZ (Fig. S12). Overall, the quenching effects were not satisfied for each fluorescent component with the relative high decrease of the fluorescent intensity observed for C3 in BD3 systems and C6 in BD5 systems. Specifically, the average decreasing percentage of the fluorescence intensity of C3 and C6 at CAP/CMZ concentration of 10 mg L^{-1} were 59.96% and 71.73%, respectively; the counterparts of CAP and CMZ were 71.12% and 60.57%. These results were confirmed by comparison of the quenching effects among the fluorescent components at each CAP/CMZ concentration (Fig. 8A). That is, a significant difference (low fluorescent intensity) was found for C3 and C6 from other fluorescent components in BD3 and BD5 systems in addition to the BD5-CMZ system. For the BD5-CMZ system, it could be seen that the apparent stronger quenching effects at CAP/CMZ concentration of 10 mg L^{-1} were related to C6 (the dashed circle in BD5-CMZ of Fig. 8A), although the overall fluorescent intensity of C7 was low. As the above analysis, C3 and C6 are both protein-like substances, revealing that the strong binding behavior might occur between the pharmaceuticals and the protein-like substance of BDOM despite the pyrolysis temperatures and the types of the drugs. The previous studies also confirmed that protein-like substances bind more strongly with the selected drugs than humic-like substances (Huang et al., 2022; Yuan et al., 2019).

In order to visually present the binding affinity in different binding systems, we pairwise compared the quenching effects between CAP and CMZ, 300°C and 500°C , and inside and outside of the bags with different molecular weights (Fig. 8B). Because the strong quenching effect was only found for C3 and C6, the data in BD3 and BD5 referred to C3 and C6. As Fig. 8B shows, the data points of BD3 (the orange square dots) and CAP (the red circle dots) were distributed below the diagonal; while the data points of BD5 (the blue square dots) and CMZ (the green circle dots) were distributed above the diagonal line. It indicates that the binding affinity between BD3 and CMZ and BD5 and CAP was relatively high. It manifested that the protein-like materials in BDOM pyrolyzed at different temperatures showed different binding affinity towards

different drugs. However, no significant difference was observed when comparing the binding affinity of BDOM in and outside of the dialysis bags towards the two drugs. Combining the results in dialysis equilibrium and fluorescent quenching, it could be known that the protein-like was the dominant component in the small molecular weight of BDOM (outside of the bag) which was strongly bound with the drugs. The BDOM solution inside the bags contained both the large and small molecular weights of BDOM, hence the BDOM in and outside of the bags showed similar binding affinity to the pharmaceuticals. It was noteworthy that more data points were displayed above the diagonal line with increasing the molecular weight of dialysis bags, illustrating that a high binding affinity to the pharmaceuticals was with respect to the BDOM inside of the bags for the larger molecular weight of the bags. The reason for this phenomenon might be that more larger molecular weight of BDOM with a humic-like substance is transferred to outside of the larger cut-off molecular weight dialysis bags which would reduce the binding affinity of the BDOM at a certain concentration (i.e., 10 mg L^{-1}).

In order to quantify the binding ability of CAP and CMZ to the fluorescent components of BDOM, the Ryan-Weber model was introduced in this study. Due to the weak quenching effect of the other fluorophores, the $\log K_M$ values were calculated only for C3 and C6. As Table 1 shows, the average $\log K_M$ values of CAP and CMZ were 4.00 ± 0.53 and 4.07 ± 0.68 , respectively; while the counterparts of C3 and C6 were 3.83 ± 0.51 and 4.51 ± 0.52 , and of BDOM inside and outside the bags were 4.19 ± 0.76 and 4.13 ± 0.54 , respectively. The $\log K_M$ values of BDOM in this study were comparable to those values of CAP or CMZ binding to the natural DOM (Huang et al., 2022; Wang et al., 2016; Yuan et al., 2019) but one order of magnitude higher than the corresponding values between soil DOM and sulfadiazine (Yuan et al., 2023), revealing that the binding behavior between BDOM and the pharmaceuticals was similar to the natural DOM and the binding affinity was related to the different types of drugs. Moreover, the apparent high values of $\log K_M$ were observed for BD5 compared to BD3 (Fig. S13), indicating that the BDOM pyrolyzed at higher temperatures showed higher binding affinity towards the pharmaceuticals. This phenomenon was also found for the different sources of BDOM binding with copper (Huang et al., 2021; Yan et al., 2023), which might be attributed to the loss of aromatic materials and oxygen-containing functional groups from BDOM as the pyrolysis temperature increased.

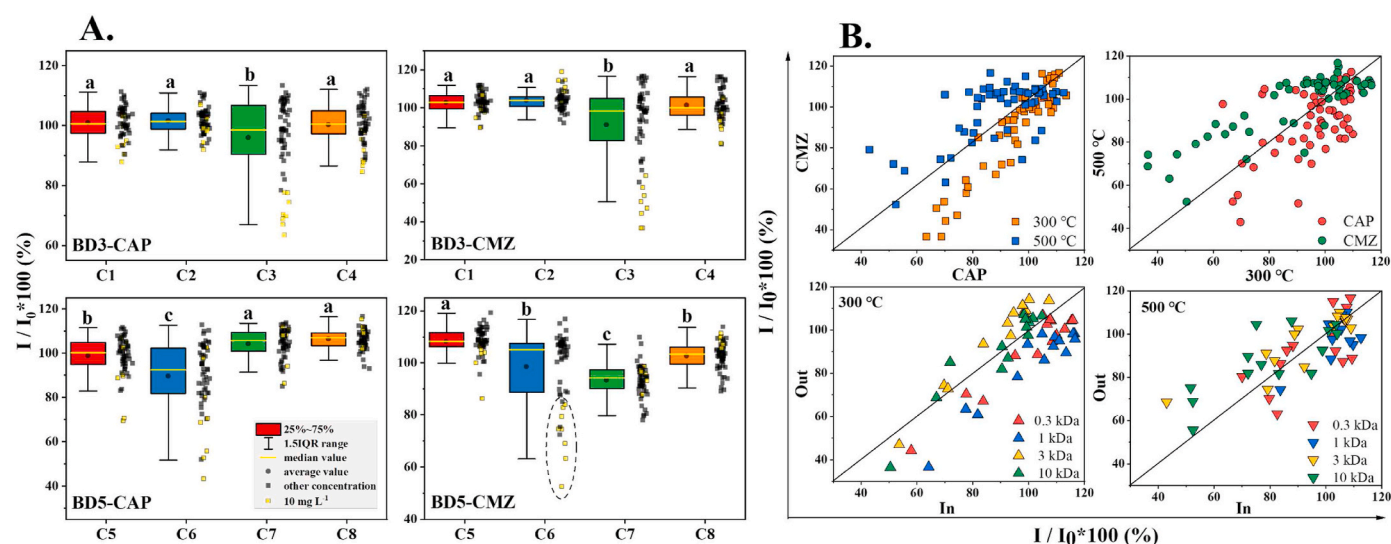


Fig. 8. Percentage of the fluorescent intensity between CAP/CMZ concentration (I) and the initial concentration (I_0) (The letter presents the significant difference test, $p < 0.05$) (A.) and the comparisons of the former percentage (I/I_0) between CAP and CMZ, 300°C and 500°C , and inside and outside of bags (B.).

Table 1
Binding parameters for the components of BDOM with CAP/CMZ.

Samples	Parameters	C3 (300 °C)		Samples	Parameters	C6 (500 °C)	
		CAP	CMZ			CAP	CMZ
BD3-0.3 I	LogK _M	4.18	3.87	BD5-0.3 I	LogK _M	4.60	4.95
	R ²	0.94	1.00		R ²	0.72	0.76
BD3-0.3 O	LogK _M	3.95	3.48	BD5-0.3 O	LogK _M	3.78	4.28
	R ²	0.95	1.00		R ²	0.84	0.91
BD3-1 I	LogK _M	4.21	4.00	BD5-1 I	LogK _M	4.90	5.89
	R ²	0.88	0.95		R ²	0.59	0.93
BD3-1 O	LogK _M	5.00	3.40	BD5-1 O	LogK _M	–	3.95
	R ²	0.99	1.00		R ²	–	0.96
BD3-3 I	LogK _M	2.93	3.30	BD5-3 I	LogK _M	4.65	3.74
	R ²	1.00	0.99		R ²	0.88	0.96
BD3-3 O	LogK _M	3.95	3.30	BD5-3 O	LogK _M	3.98	4.36
	R ²	0.98	1.00		R ²	0.94	0.91
BD3-10 I	LogK _M	3.85	3.11	BD5-10 I	LogK _M	4.99	3.89
	R ²	0.92	1.00		R ²	0.73	0.90
BD3-10 O	LogK _M	3.95	4.89	BD5-10 O	LogK _M	4.97	4.72
	R ²	0.98	0.94		R ²	0.89	0.96

“–”: failed to be modeled.

3.4. Environmental Implication

This study first investigated the binding mechanism of BDOM from an invasive plant and drugs at different pyrolysis temperatures based on multiple spectroscopic techniques. It was found that pyrolysis temperature had a greater effect on BDOM-drug binding than BDOM molecular weight, which offers recommendations for future biochar application in the environment. In addition, the exploration of the binding site, binding substance, binding order, and binding affinity revealed that the protein-like substance strongly impacts the BDOM-drug binding behavior. Besides, our results are also conducive to deeply assessing contaminant fate and the risk of extensive biochar application on ecosystems.

4. Conclusion

In this study, the properties of BDOM derived from *Alternanthera philoxeroides* at different pyrolysis temperatures and the effects of pyrolysis temperature and molecular weight on the combination of BDOM with drugs were investigated using various spectral techniques and methods. Generally, the humic-like substance was the dominant component for all the BDOM regardless of the pyrolysis temperature. As pyrolysis temperature increased, a smoother and more rigid fibrous structure, lower DOC concentration, and the carboxyl groups on the BDOM were gradually replaced by the aliphatic chains. Besides, the relatively higher binding capacity between the pharmaceuticals and BDOM was found for BD5 and BD-0.3. In addition, the protein-like substance not only showed more susceptible and released precedingly during the dialysis equilibrium process but also greatly impacted the binding capacity of BDOM-CAP&CMZ and was more susceptibly bound with the pharmaceuticals. In particular, the aliphatic C–OH, amide II N–H, carboxyl C=O, and phenolic-OH in the tryptophan-like substance preferred binding to the pharmaceuticals. Besides, a significantly higher binding affinity towards the pharmaceuticals was related to the BDOM pyrolyzed at higher temperatures (500 °C). However, no apparent difference was observed for the binding affinity of BDOM with different molecular weights.

CRediT authorship contribution statement

Wangyu Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Minghua Nie:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis,

Conceptualization. **Caixia Yan:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Yulong Yuan:** Writing – review & editing, Methodology, Data curation. **Aoxue Xu:** Writing – review & editing, Methodology. **Mingjun Ding:** Writing – review & editing, Validation. **Peng Wang:** Writing – review & editing, Methodology. **Min Ju:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (42067058 and 42067034). Additional funding was provided by the Jiangxi Provincial Natural Science Foundation (20232ACB213014, 20232BAB203083, and 20212BCJL23058).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2024.123867>.

References

Bani-Yaseen, A.D., 2011. Spectrofluorimetric study on the interaction between antimicrobial drug sulfamethazine and bovine serum albumin. *J. Lumin.* 131, 1042–1047.
Bi, H., Tang, L., Gao, X., Jia, J., Lv, H., 2016. Spectroscopic analysis on the binding interaction between tetracycline hydrochloride and bovine proteins β-casein, α-lactalbumin. *J. Lumin.* 178, 72–83.
Catalá, T.S., Reche, I., Fuentes-Lema, A., Romera-Castillo, C., Nieto-Cid, M., Ortega-Retuerta, E., Calvo, E., Alvarez, M., Marrasé, C., Stedmon, C.A., 2015. Turnover time of fluorescent dissolved organic matter in the dark global ocean. *Nat. Commun.* 6, 5986.
Catalán, N., Pastor, A., Borrego, C.M., Casas-Ruiz, J.P., Hawkes, J.A., Gutiérrez, C., von Schiller, D., Marcé, R., 2021. The relevance of environment vs. composition on dissolved organic matter degradation in freshwaters. *Limnol. Oceanogr.* 66, 306–320.

- Cheng, J., Hu, S., Sun, G., Geng, Z., Zhu, M., 2021. The effect of pyrolysis temperature on the characteristics of biochar, pyrolytic acids, and gas prepared from cotton stalk through a polygeneration process. *Ind. Crop. Prod.* 170, 113690.
- Coble, P.G., 1996. Characterization of marine and terrestrial DOM in seawater using excitation-emission matrix spectroscopy. *Mar. Chem.* 51, 325–346.
- Dong, X., Ma, L.Q., Gress, J., Harris, W., Li, Y., 2014. Enhanced Cr (VI) reduction and as (III) oxidation in ice phase: important role of dissolved organic matter from biochar. *J. Hazard Mater.* 267, 62–70.
- Fan, T., Yao, X., Ren, H., Liu, L., Deng, H., Shao, K., 2022. Regional-scale investigation of the molecular weight distribution and metal-binding behavior of dissolved organic matter from a shallow macrophytic lake using multispectral techniques. *J. Hazard Mater.* 439, 129532.
- Fan, T., Yao, X., Sun, Z., Sang, D., Liu, L., Deng, H., Zhang, Y., 2023. Properties and metal binding behaviors of sediment dissolved organic matter (SDOM) in lakes with different trophic states along the Yangtze River Basin: a comparison and summary. *Water Res.* 231, 119605.
- Feng, Q., Wang, B., Chen, M., Wu, P., Lee, X., Xing, Y., 2021. Invasive plants as potential sustainable feedstocks for biochar production and multiple applications: a review. *Resour. Conserv. Recycl.* 164, 105204.
- Fuentes, M., González-Gaitano, G., García-Mina, J.M., 2006. The usefulness of UV-visible and fluorescence spectroscopies to study the chemical nature of humic substances from soils and composts. *Org. Geochem.* 37, 1949–1959.
- Gui, X., Liu, C., Li, F., Wang, J., 2020. Effect of pyrolysis temperature on the composition of DOM in manure-derived biochar. *Ecotoxicol. Environ. Saf.* 197, 110597.
- Gullian-Klanian, M., Gold-Bouchot, G., Delgadillo-Díaz, M., Aranda, J., Sánchez-Solís, M. J., 2021. Effect of the use of *Bacillus* spp. on the characteristics of dissolved fluorescent organic matter and the phytochemical quality of *Stevia rebaudiana* grown in a recirculating aquaponic system. *Environ. Sci. Pollut. Control Ser.* 28, 36326–36343.
- Guo, X., Peng, Y., Li, N., Tian, Y., Dai, L., Wu, Y., Huang, Y., 2022. Effect of biochar-derived DOM on the interaction between Cu (II) and biochar prepared at different pyrolysis temperatures. *J. Hazard Mater.* 421, 126739.
- Hao, S., Zhu, X., Liu, Y., Qian, F., Fang, Z., Shi, Q., Zhang, S., Chen, J., Ren, Z.J., 2018. Production temperature effects on the structure of hydrochar-derived dissolved organic matter and associated toxicity. *Environmental Science & Technology* 52, 7486–7495.
- He, C., He, X., Li, J., Luo, Y., Li, J., Pei, Y., Jiang, J., 2021a. The spectral characteristics of biochar-derived dissolved organic matter at different pyrolysis temperatures. *J. Environ. Chem. Eng.* 9, 106075.
- He, M., Xiong, X., Wang, L., Hou, D., Bolan, N.S., Ok, Y.S., Rinklebe, J., Tsang, D.C., 2021b. A critical review on performance indicators for evaluating soil biota and soil health of biochar-amended soils. *J. Hazard Mater.* 414, 125378.
- Hong, H., Wu, S., Wang, Q., Dai, M., Qian, L., Zhu, H., Li, J., Zhang, J., Liu, J., Li, J., 2021. Fluorescent dissolved organic matter facilitates the phytoavailability of copper in the coastal wetlands influenced by artificial topography. *Sci. Total Environ.* 790, 147855.
- Huang, M., Li, Z., Chen, M., Wen, J., Luo, N., Xu, W., Ding, X., Xing, W., 2020. Dissolved organic matter released from rice straw and straw biochar: Contrasting molecular composition and lead binding behaviors. *Sci. Total Environ.* 739, 140378.
- Huang, M., Li, Z., Luo, N., Yang, R., Wen, J., Huang, B., Zeng, G., 2019. Application potential of biochar in environment: Insight from degradation of biochar-derived DOM and complexation of DOM with heavy metals. *Sci. Total Environ.* 646, 220–228.
- Huang, M., Li, Z., Wen, J., Ding, X., Zhou, M., Cai, C., Shen, F., 2021. Molecular insights into the effects of pyrolysis temperature on composition and copper binding properties of biochar-derived dissolved organic matter. *J. Hazard Mater.* 410, 124537.
- Huang, X., Liang, Y., Ye, Q., Ding, Z., Liu, F., Shi, Z., 2023. Theoretical modeling of molecular fractionation of dissolved organic matter on ferrihydrite and its impact on proton and metal binding properties. *Sci. Total Environ.* 888, 164276.
- Huang, X., Yan, C., Nie, M., Chen, J., Ding, M., 2022. Effect of colloidal fluorescence properties on the complexation of chloramphenicol and carbamazepine to the natural aquatic colloids. *Chemosphere* 286, 131604.
- Ihsanullah, I., Khan, M.T., Zubair, M., Bilal, M., Sajid, M., 2022. Removal of pharmaceuticals from water using sewage sludge-derived biochar: a review. *Chemosphere* 289, 133196.
- Jaffé, R., Ding, Y., Niggemann, J., Vähätalo, A.V., Stubbins, A., Spencer, R.G., Campbell, J., Dittmar, T., 2013. Global charcoal mobilization from soils via dissolution and riverine transport to the oceans. *Science* 340, 345–347.
- Jiang, S., Dai, G., Liu, Z., He, T., Zhong, J., Ma, Y., Shu, Y., 2022. Field-scale fluorescence fingerprints of biochar-derived dissolved organic matter (DOM) provide an effective way to trace biochar migration and the downward co-migration of Pb, Cu and as in soil. *Chemosphere* 301, 134738.
- Joseph, S.D., Camps-Arbestain, M., Lin, Y., Munroe, P., Chia, C., Hook, J., Van Zwieten, L., Kimber, S., Cowie, A., Singh, B., 2010. An investigation into the reactions of biochar in soil. *Soil Res.* 48, 501–515.
- Lambert, T., Bouillon, S., Darchambeau, F., Massicotte, P., Borges, A.V., 2016. Shift in the chemical composition of dissolved organic matter in the Congo River network. *Biogeosciences* 13, 5405–5420.
- Lambert, T., Bouillon, S., Darchambeau, F., Morana, C., Roland, F.A.E., Descy, J.P., Borges, A.V., 2017. Effects of human land use on the terrestrial and aquatic sources of fluvial organic matter in a temperate river basin (The Meuse River, Belgium). *Biogeochemistry* 136, 191–211.
- Li, L.-P., Liu, Y.-H., Ren, D., Wang, J.-J., 2022. Characteristics and chlorine reactivity of biochar-derived dissolved organic matter: effects of feedstock type and pyrolysis temperature. *Water Res.* 211, 118044.
- Lian, W., Yang, L., Joseph, S., Shi, W., Bian, R., Zheng, J., Li, L., Shan, S., Pan, G., 2020. Utilization of biochar produced from invasive plant species to efficiently adsorb Cd (II) and Pb (II). *Bioresour. Technol.* 317, 124011.
- Lin, H., Guo, L.D., 2020. Variations in colloidal DOM composition with molecular weight within individual water samples as characterized by flow field-flow fractionation and EEM-PARAFAC analysis. *Environmental Science & Technology* 54, 1657–1667.
- Liu, C.-H., Chu, W., Li, H., Boyd, S.A., Teppen, B.J., Mao, J., Lehmann, J., Zhang, W., 2019. Quantification and characterization of dissolved organic carbon from biochars. *Geoderma* 335, 161–169.
- Liu, D., Gao, H., Yu, H., Song, Y., 2022. Applying EEM-PARAFAC combined with moving-window 2DCOS and structural equation modeling to characterize binding properties of Cu (II) with DOM from different sources in an urbanized river. *Water Res.* 227, 119317.
- Liu, P., Pateck, C.J., Blowes, D.W., Berti, W.R., Landis, R.C., 2015. Aqueous leaching of organic acids and dissolved organic carbon from various biochars prepared at different temperatures. *J. Environ. Qual.* 44, 684–695.
- Liu, Y., Zhou, S., Fu, Y., Sun, X., Li, T., Yang, C., 2023. Characterization of dissolved organic matter in biochar derived from various macroalgae (Phaeophyta, Rhodophyta, and Chlorophyta): effects of pyrolysis temperature and extraction solution pH. *Sci. Total Environ.* 869, 161786.
- Murphy, K.R., Hambly, A., Singh, S., Henderson, R.K., Baker, A., Stuetz, R., Khan, S.J., 2011. Organic matter fluorescence in municipal water recycling schemes: toward a unified PARAFAC model. *Environmental Science & Technology* 45, 2909–2916.
- Murphy, K.R., Stedmon, C.A., Wenig, P., Bro, R., 2014. OpenFluor—an online spectral library of auto-fluorescence by organic compounds in the environment. *Anal. Methods* 6, 658–661.
- Murphy, K.R., Timko, S.A., Gonsior, M., Powers, L.C., Wunsch, U.J., Stedmon, C.A., 2018. Photochemistry illuminates ubiquitous organic matter fluorescence spectra. *Environmental Science & Technology* 52, 11243–11250.
- Noda, I., 1990. Two-dimensional infrared (2D IR) spectroscopy: theory and applications. *Appl. Spectrosc.* 44, 550–561.
- Noda, I., 2006. Progress in two-dimensional (2D) correlation spectroscopy. *J. Mol. Struct.* 799, 2–15.
- Özçimen, D., Ersoy-Meriçboyu, A., 2010. Characterization of biochar and bio-oil samples obtained from carbonization of various biomass materials. *Renew. Energy* 35, 1319–1324.
- Paini, D.R., Sheppard, A.W., Cook, D.C., De Barro, P.J., Worner, S.P., Thomas, M.B., 2016. Global threat to agriculture from invasive species. *Proc. Natl. Acad. Sci. USA* 113, 7575–7579.
- Poerschmann, J., Weiner, B., Wedwitschka, H., Zehndorf, A., Koehler, R., Kopinke, F.-D., 2015. Characterization of biochars and dissolved organic matter phases obtained upon hydrothermal carbonization of *Elodea nuttallii*. *Bioresour. Technol.* 189, 145–153.
- Rajapaksha, A.U., Ok, Y.S., El-Naggar, A., Kim, H., Song, F., Kang, S., Tsang, Y.F., 2019. Dissolved organic matter characterization of biochars produced from different feedstock materials. *J. Environ. Manag.* 233, 393–399.
- Rizzuto, S., Jones, K.C., Zhang, H., Baho, D.L., Leu, E., Nizzetto, L., 2021. Critical assessment of an equilibrium-based method to study the binding of waterborne organic contaminants to natural dissolved organic matter (DOM). *Chemosphere* 285, 131524.
- Romera-Castillo, C., Chen, M., Yamashita, Y., Jaffé, R., 2014. Fluorescence characteristics of size-fractionated dissolved organic matter: implications for a molecular assembly based structure? *Water Res.* 55, 40–51.
- Ryan, D.K., Weber, J.H., 1982. Fluorescence quenching titration for determination of complexing capacities and stability constants of fulvic acid. *Anal. Chem.* 54, 986–990.
- Shaheen, S.M., Niazi, N.K., Hassan, N.E., Bibi, I., Wang, H., Tsang, D.C., Ok, Y.S., Bolan, N., Rinklebe, J., 2019. Wood-based biochar for the removal of potentially toxic elements in water and wastewater: a critical review. *Int. Mater. Rev.* 64, 216–247.
- Spokas, K., Novak, J., Masiello, C., Johnson, M., Colosky, E., Ippolito, J., Trigo, C., 2014. Physical disintegration of biochar: an overlooked process. *Environ. Sci. Technol. Lett.* 1, 326–332.
- Sun, Y., Xiong, X., He, M., Xu, Z., Hou, D., Zhang, W., Ok, Y.S., Rinklebe, J., Wang, L., Tsang, D.C., 2021. Roles of biochar-derived dissolved organic matter in soil amendment and environmental remediation: a critical review. *Chem. Eng. J.* 424, 130387.
- Wang, B., Li, W., Wang, J., 2005. Genetic diversity of *Alternanthera philoxeroides* in China. *Aquat. Bot.* 81, 277–283.
- Wang, J., Lu, X., Jing, Q., Zhang, B., Ye, J., Zhang, H., Xiao, Z., Zhang, J., 2023. Spatiotemporal characterization of heavy metal and antibiotics in the Pearl River Basin and pollutants removal assessment using invasive species-derived biochars. *J. Hazard Mater.* 454, 131409.
- Wang, K., Sun, Y., Tang, J., He, J., Sun, H., 2020a. Aqueous Cr (VI) removal by a novel ball milled Fe⁰-biochar composite: role of biochar electron transfer capacity under high pyrolysis temperature. *Chemosphere* 241, 125044.
- Wang, L., O'Connor, D., Rinklebe, J., Ok, Y.S., Tsang, D.C., Shen, Z., Hou, D., 2020b. Biochar aging: Mechanisms, physicochemical changes, assessment, and implications for field applications. *Environmental Science & Technology* 54, 14797–14814.
- Wang, Y., Zhang, M., Fu, J., Li, T., Wang, J., Fu, Y., 2016. Insights into the interaction between carbamazepine and natural dissolved organic matter in the Yangtze Estuary using fluorescence excitation–emission matrix spectra coupled with parallel factor analysis. *Environ. Sci. Pollut. Control Ser.* 23, 19887–19896.
- Wei, S., Zhu, M., Fan, X., Song, J., Li, K., Jia, W., Song, H., 2019. Influence of pyrolysis temperature and feedstock on carbon fractions of biochar produced from pyrolysis of rice straw, pine wood, pig manure and sewage sludge. *Chemosphere* 218, 624–631.

- Wu, J., Tu, W., Li, C., He, F., 2022. Binding characteristics of copper onto biochar-derived DOM using general, heterospectral and moving-window two-dimensional correlation analyses. *J. Hazard Mater.* 435, 129021.
- Xu, H., Yan, M., Li, W., Jiang, H., Guo, L., 2018. Dissolved organic matter binding with Pb (II) as characterized by differential spectra and 2D UV-FTIR heterospectral correlation analysis. *Water Res.* 144, 435–443.
- Yamashita, Y., Panton, A., Mahaffey, C., Jaffé, R., 2011. Assessing the spatial and temporal variability of dissolved organic matter in Liverpool Bay using excitation-emission matrix fluorescence and parallel factor analysis. *Ocean Dynam.* 61, 569–579.
- Yan, C., Liu, H., Sheng, Y., Huang, X., Nie, M., Huang, Q., Baalousha, M., 2018. Fluorescence characterization of fractionated dissolved organic matter in the five tributaries of Poyang Lake, China. *Sci. Total Environ.* 637–638, 1311–1320.
- Yan, C., Nie, M., Yang, Y., Zhou, J., Liu, M., Baalousha, M., Lead, J.R., 2015. Effect of colloids on the occurrence, distribution and photolysis of emerging organic contaminants in wastewaters. *J. Hazard Mater.* 299, 241–248.
- Yan, C., Wang, W., Nie, M., Ding, M., Wang, P., Zhang, H., Huang, G., 2023. Characterization of copper binding to biochar-derived dissolved organic matter: effects of pyrolysis temperature and natural wetland plants. *J. Hazard Mater.* 442, 130076.
- Yan, C., Wang, X., Nie, M., Mo, X., Ding, M., Chen, J., Yang, Y., 2024. Characteristics of microplastic-derived dissolved organic matter and its binding with pharmaceuticals unveiled by fluorescence spectroscopy and two-dimensional correlation spectroscopy. *Sci. Total Environ.* 908, 168190.
- Yang, F., Xue, Y., Gao, Y., Zhu, Q., Wang, C., Sun, H., 2023. Biochar-derived dissolved organic matters influencing bacterium characteristics during biodegradation of sulfamethoxazole and chloramphenicol under alternation of visible and avoiding light. *Biochar* 5, 9.
- Yang, F., Zhang, Q., Jian, H., Wang, C., Hao, Y., 2020a. Effect of biochar-derived dissolved organic matter on adsorption of sulfamethoxazole and chloramphenicol. *J. Hazard Mater.* 396, 122598.
- Yang, F., Zhang, Q., Jian, H., Wang, C., Xing, B., Sun, H., Hao, Y., 2020b. Effect of biochar-derived dissolved organic matter on adsorption of sulfamethoxazole and chloramphenicol. *J. Hazard Mater.* 396, 122598.
- Ye, W., Li, J., Cao, H., Ge, X., 2003. Genetic uniformity of *Alternanthera philoxeroides* in South China. *Weed Res.* 43, 297–302.
- Yu, S., Zhang, H., Ni, J., Xiang, Y., Wei, R., Qian, W., Chen, W., 2023. Spectral characteristics coupled with self-organizing maps analysis on different molecular size-fractionated water-soluble organic carbon from biochar. *Sci. Total Environ.* 857, 159424.
- Yuan, D., Wang, H., An, Y., Guo, X., He, L., 2019. Insight into the binding properties of carbamazepine onto dissolved organic matter using spectroscopic techniques during grassy swale treatment. *Ecotoxicol. Environ. Saf.* 173, 444–451.
- Yuan, D., Xiong, S., Yan, C., Zhai, L., Cui, Y., Kou, Y., 2023. Simultaneous degradation of sulfadiazine and dissolved organic matter based on low-impact development facilities. *Journal of Environmental Sciences* 130, 223–233.
- Zhang, H., Wu, L., Qian, W., Ni, J., Wei, R., Qi, Z., Chen, W., 2021a. Spectral characteristics of dissolved organic carbon derived from biomass-pyrogenic smoke (SDOC) in the aqueous environment and its solubilization effect on hydrophobic organic pollutants. *Water Res.* 203, 117515.
- Zhang, J., Zhou, S., Sun, H., Lü, F., He, P., 2020a. The soluble fraction from straw-derived biochar supplies nutrients and affects carbon storage of coastal mudflat soil in rice paddy. *Environ. Sci. Pollut. Control Ser.* 27, 18079–18088.
- Zhang, P., Duan, W., Peng, H., Pan, B., Xing, B., 2021b. Functional biochar and its balanced design. *ACS Environmental Au* 2, 115–127.
- Zhang, P., Huang, P., Xu, X., Sun, H., Jiang, B., Liao, Y., 2020b. Spectroscopic and molecular characterization of biochar-derived dissolved organic matter and the associations with soil microbial responses. *Sci. Total Environ.* 708, 134619.
- Zhang, P., Meng, X., Liu, A., Ma, M., Shao, Y., Sun, H., 2023. Biochar-derived dissolved black carbon accelerates ferrihydrite microbial transformation and subsequent imidacloprid degradation. *J. Hazard Mater.* 446, 130685.
- Zhou, X., Lai, C., Almatrafi, E., Liu, S., Yan, H., Qian, S., Li, H., Qin, L., Yi, H., Fu, Y., 2023. Unveiling the roles of dissolved organic matters derived from different biochar in biochar/persulfate system: mechanism and toxicity. *Sci. Total Environ.* 864, 161062.